

## 13.2 Scheffé's projection method

Scheffé's method answers the hardest version of the problem: the family is every estimable function in a subspace, and the analyst may choose which ones to look at after seeing the data. The intervals have simultaneous coverage exactly  $1 - \alpha$ , because the largest standardized error in the family is the length of a projection, and Scheffé's method *is* the  $F$  test, read one direction at a time.

Throughout this section the model is the normal linear model Eq. 7.3.1,  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , with  $r = \text{rank}(\mathbf{X}) < n$ . Write  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ ,  $\nu = n - r$  for the error degrees of freedom,  $s^2 = \text{SSE}/\nu$ ,  $\mathbf{G} = (\mathbf{X}^\top\mathbf{X})^-$  for any generalized inverse, and  $F_\alpha(q, \nu)$  for the upper  $\alpha$  point of  $F(q, \nu)$ . For an estimable  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$  the standard error is  $\text{se}(\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda}}$  (Corollary 7.5.2).

### 13.2.1 Linear functions of the mean as vectors

An estimable function is a linear function of the mean vector, and a linear function of  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  can be represented by a vector of  $\mathcal{C}(\mathbf{X})$ . If  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$  is estimable, then  $\boldsymbol{\lambda} = \mathbf{X}^\top\boldsymbol{\rho}$  for some vector  $\boldsymbol{\rho} \in \mathbb{R}^n$  (Theorem 8.3.1), and with  $\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$ ,

$$\boldsymbol{\lambda}^\top\boldsymbol{\beta} = \boldsymbol{\rho}^\top\mathbf{X}\boldsymbol{\beta} = \mathbf{v}^\top\boldsymbol{\mu}, \quad \boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Y} = \mathbf{v}^\top\mathbf{Y}, \quad \boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\rho}^\top\mathbf{X}\mathbf{G}\mathbf{X}^\top\boldsymbol{\rho} = \boldsymbol{\rho}^\top\mathbf{M}\boldsymbol{\rho} = \|\mathbf{v}\|^2. \quad (13.2.1)$$

The second identity is Theorem 6.4.4, and the third uses  $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$  (Theorem 6.3.6). The vector  $\mathbf{v}$  does not depend on the choice of  $\boldsymbol{\rho}$ , since two choices differ by some  $\boldsymbol{\delta}$  with  $\mathbf{X}^\top\boldsymbol{\delta} = \mathbf{0}$ , hence  $\mathbf{M}\boldsymbol{\delta} = \mathbf{0}$ . So the estimate is the inner product of  $\mathbf{v}$  with the data, the estimand its inner product with the mean, and the standard error is  $s\|\mathbf{v}\|$ .

A *family* of estimable functions closed under linear combination is then a subspace  $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$ . It is usually described in one of two ways. If the family is  $\{\boldsymbol{\lambda}^\top\boldsymbol{\beta} : \boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})\}$  for a  $p \times q$  matrix  $\boldsymbol{\Lambda} = \mathbf{X}^\top\mathbf{T}$ , then  $\mathbf{v} = \mathbf{M}\mathbf{T}\mathbf{a}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{M}\mathbf{T})$ , the test space of  $\boldsymbol{\Lambda}^\top\boldsymbol{\beta} = \mathbf{0}$  (Theorem 6.5.2); if  $\boldsymbol{\Lambda}$  has rank  $q$ , so does  $\mathbf{M}\mathbf{T}$ , because  $\mathbf{X}^\top\mathbf{M}\mathbf{T} = \boldsymbol{\Lambda}$ . If instead the family is the set of functions about which a reduced model  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  makes a claim, those vanishing on  $\mathcal{C}(\mathbf{X}_0)$ , then  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ , of dimension  $r - r_0$  (Theorem 6.5.1).

### 13.2.2 The theorem

#### Lemma 13.2.1. The largest standardized component

Let  $\mathcal{V}$  be a subspace of  $\mathbb{R}^n$  with orthogonal projection  $\mathbf{P}_\mathcal{V}$ , and let  $\mathbf{u} \in \mathbb{R}^n$ . Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top\mathbf{u})^2}{\|\mathbf{v}\|^2} = \|\mathbf{P}_\mathcal{V}\mathbf{u}\|^2,$$

and if  $\mathbf{P}_\mathcal{V}\mathbf{u} \neq \mathbf{0}$  the maximum is attained exactly at the nonzero multiples of  $\mathbf{P}_\mathcal{V}\mathbf{u}$ .

*Proof.* For  $\mathbf{v} \in \mathcal{V}$ ,  $\mathbf{v} = \mathbf{P}_\mathcal{V}\mathbf{v}$ , and since  $\mathbf{P}_\mathcal{V}$  is symmetric,  $\mathbf{v}^\top\mathbf{u} = \mathbf{v}^\top\mathbf{P}_\mathcal{V}\mathbf{u}$ . By Cauchy–Schwarz (Proposition 1.1.1),  $(\mathbf{v}^\top\mathbf{P}_\mathcal{V}\mathbf{u})^2 \leq \|\mathbf{v}\|^2\|\mathbf{P}_\mathcal{V}\mathbf{u}\|^2$ , with equality iff  $\mathbf{v}$  is a multiple of  $\mathbf{P}_\mathcal{V}\mathbf{u}$ . That

multiple lies in  $\mathcal{V}$ , so the bound is attained. If  $\mathbf{P}_{\mathcal{V}}\mathbf{u} = \mathbf{0}$  both sides are zero. ■

The lemma is Exercise (C1) of [Chapter 9](#), where it described the sum of squares of a row of an analysis of variance table as the largest one-degree-of-freedom sum of squares inside it. Applied to  $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$ , it describes the worst standardized error in a family.

### Theorem 13.2.2. Scheffé's simultaneous intervals

Assume Eq. 7.3.1 with  $r < n$ , and let  $0 < \alpha < 1$ .

- (a) Let  $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$  be a subspace of dimension  $q \geq 1$ . Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu})^2}{q s^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{P}_{\mathcal{V}}(\mathbf{Y} - \boldsymbol{\mu})\|^2 / q}{s^2} \sim F(q, \nu),$$

and consequently

$$\Pr\left(|\mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})| \leq \sqrt{q F_\alpha(q, \nu)} s \|\mathbf{v}\| \quad \forall \mathbf{v} \in \mathcal{V}\right) = 1 - \alpha.$$

- (b) Let  $\boldsymbol{\Lambda}$  be  $p \times q$  of rank  $q$  with  $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . With probability exactly  $1 - \alpha$ , for every  $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$  simultaneously,

$$\boldsymbol{\lambda}^\top \boldsymbol{\beta} \in \left[ \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \sqrt{q F_\alpha(q, \nu)} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}), \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \sqrt{q F_\alpha(q, \nu)} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \right]. \quad (13.2.2)$$

Moreover, with  $\boldsymbol{\theta} = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ ,  $\hat{\boldsymbol{\theta}} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$  and  $\mathbf{W} = \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda}$ ,

$$\max_{\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda}), \boldsymbol{\lambda} \neq \mathbf{0}} \frac{(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})^2}{\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2} = \frac{(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})}{s^2}, \quad (13.2.3)$$

attained at  $\boldsymbol{\lambda} = \boldsymbol{\Lambda} \mathbf{W}^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})$ .

*Proof.* (a) The first equality is [Lemma 13.2.1](#) with  $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$ . The map  $\mathbf{v} \mapsto \mathbf{X}^\top \mathbf{v}$  is one to one on  $\mathcal{C}(\mathbf{X})$  (Step 1 of the proof of Theorem 12.4.1), so  $\mathcal{L} = \{\mathbf{X}^\top \mathbf{v} : \mathbf{v} \in \mathcal{V}\}$  is a subspace of  $\mathcal{C}(\mathbf{X}^\top)$  of dimension  $q$ . For  $\mathbf{x} = \mathbf{X}^\top \mathbf{v}$ , [Eq. 13.2.1](#) with  $\boldsymbol{\rho} = \mathbf{v}$  gives  $\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta} = \mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})$  and  $\mathbf{x}^\top \mathbf{G} \mathbf{x} = \|\mathbf{v}\|^2$ . So the event in the display is the event Eq. 12.4.1 with  $k = q$ , and it has probability exactly  $1 - \alpha$  by Theorem 12.4.1. That event is also the event that the maximum is at most  $F_\alpha(q, \nu)$ . Since this holds for every  $\alpha \in (0, 1)$ , the maximum has the distribution function of  $F(q, \nu)$ .

(b) Write  $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{M} \mathbf{T})$ , of dimension  $q$ . Every  $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$  is  $\boldsymbol{\Lambda} \mathbf{a} = \mathbf{X}^\top (\mathbf{T} \mathbf{a})$ , and by [Eq. 13.2.1](#)  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu}$  and  $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s \|\mathbf{v}\|$  with  $\mathbf{v} = \mathbf{M} \mathbf{T} \mathbf{a} \in \mathcal{V}$ . As  $\mathbf{a}$  ranges over  $\mathbb{R}^q$ ,  $\mathbf{v}$  ranges over all of  $\mathcal{V}$ . So the event in (b) is the event in (a). Finally, [Eq. 13.2.3](#) is Proposition 11.6.2 for the hypothesis  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \boldsymbol{\theta}$ , which is testable because  $\boldsymbol{\theta}$  is the true value: its statistic Eq. 11.3.1 is  $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) / (q s^2)$ , and each  $T(\mathbf{a})^2$  is the ratio on the left of [Eq. 13.2.3](#) at  $\boldsymbol{\lambda} = \boldsymbol{\Lambda} \mathbf{a}$ , so the maximum is attained at  $\mathbf{a} \propto \mathbf{W}^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})$ . ■

The coverage is exact, not a bound: some function in the family, depending on the data, always

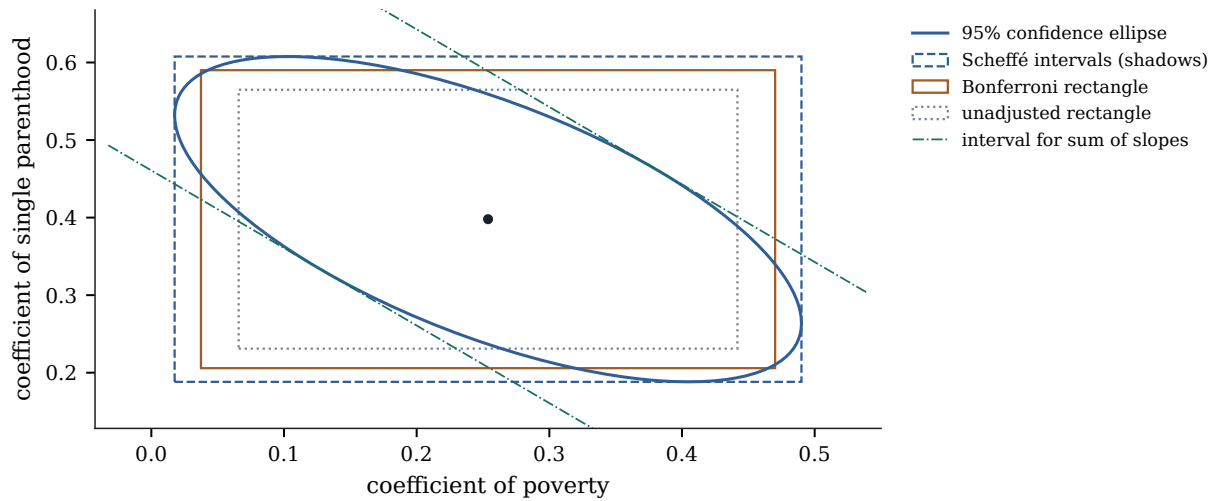


Figure 13.2.1: The coefficients of poverty and single parenthood in the crime regression. The Scheffé intervals (dashed box, multiplier 2.530) are the shadows of the 95% confidence ellipse; the Bonferroni rectangle (multiplier 2.317) covers only the two coefficients. The tangent lines bound the Scheffé interval for the sum of the slopes. The ellipse is that of Figure 12.2.1.

lies on the edge of the event. Only the subspace matters, so dependent columns of  $\mathbf{A}$  may be dropped first. For  $q = 1$  the multiplier is  $t_{\nu, \alpha/2}$ , since  $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$ , and the interval is that of Theorem 12.1.1.

Part (b) is the observation-space form of a fact met twice already. Section 11.6 proved Eq. 13.2.3, with  $\mathbf{d}$  in place of  $\boldsymbol{\theta}$ , as the statement that a group  $F$  statistic is the largest single  $t$  statistic in the group, and Section 12.2 showed that the intervals Eq. 13.2.2 are the *shadows* of the confidence ellipsoid for  $\boldsymbol{\theta} = \mathbf{A}^\top \boldsymbol{\beta}$  (Theorem 12.2.1). The ellipsoid covers  $\boldsymbol{\theta}$  iff every shadow covers the corresponding  $\mathbf{a}^\top \boldsymbol{\theta}$ , which is why an infinite family costs no more than one ellipsoid. The coverage statement itself is the Working–Hotelling band of Theorem 12.4.1 for the subspace  $\mathcal{L} = \mathcal{C}(\mathbf{A})$ : a band for a regression surface and a family of intervals for estimable functions are one object, written in the coordinates of  $\mathcal{C}(\mathbf{X}^\top)$  or of  $\mathcal{C}(\mathbf{X})$ . Figure 13.2.1 extends Figure 12.2.1, which drew the same ellipse for two slopes of the crime regression, by adding the Bonferroni rectangle and the tangent lines that bound the Scheffé interval for a combination  $a_1\beta_1 + a_2\beta_2$ .

### Example. Three slopes of a crime regression

Return to the regression of the murder rate on poverty, single parenthood and urbanization for the 50 states of Example (Poverty and murder rates), with  $\nu = 46$  error degrees of freedom. The poverty coefficient is 0.2538 with standard error 0.0934, so  $t = 2.718$ . Suppose the family of interest is every linear combination of the three slopes,  $q = 3$ . The 95% intervals for the poverty coefficient are:

| method                    | multiplier | interval        |
|---------------------------|------------|-----------------|
| unadjusted $t$            | 2.013      | [0.066, 0.442]  |
| Bonferroni, three slopes  | 2.485      | [0.022, 0.486]  |
| Scheffé, all combinations | 2.902      | [−0.017, 0.525] |

The poverty coefficient is significant on its own and among three planned comparisons, but not once every combination of the slopes is admitted. The  $F$  statistic for the hypothesis that all three slopes are zero is 27.49, and the largest  $|t|$  over all combinations is  $9.081 = \sqrt{3 \times 27.49}$ , in agreement with Eq. 13.2.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
nu = n - p                                # error degrees of freedom
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / nu
se = np.sqrt(s2 * np.diag(XtX_inv))
```

[Run this code online](#)

```
alpha, slopes = 0.05, [1, 2, 3]                # poverty, single, urban
k = len(slopes)
mult = {
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * k), nu),
    "Scheffe": np.sqrt(k * stats.f.ppf(1 - alpha, k, nu)),    # q = 3 slopes
}
for name, c in mult.items():
    lo, hi = beta_hat[slopes] - c * se[slopes], beta_hat[slopes] + c * se[slopes]
    print(f"{name:10s} c = {c:.3f} poverty [{lo[0]:.3f}, {hi[0]:.3f}]")
```

[Run this code online](#)

```
L = np.eye(p)[:, slopes]                      # Lambda: the three slopes
W = L.T @ XtX_inv @ L
d = L.T @ beta_hat
F = d @ np.linalg.solve(W, d) / (k * s2)      # F test that all slopes are 0
a_star = np.linalg.solve(W, d)                # the most significant
combination
t_star = (a_star @ d) / np.sqrt(s2 * a_star @ W @ a_star)
print(f"F = {F:.2f}; largest |t| over all combinations = {t_star:.3f} = sqrt(3F)")
```

[Run this code online](#)

### 13.2.3 Scheffé's method is the $F$ test

Let  $\mathbf{d} \in \mathbb{R}^q$  and consider the hypothesis  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ , with  $\mathbf{\Lambda}$  as in Theorem 13.2.2(b). Because  $\mathbf{\Lambda}^\top$  has rank  $q$ , the hypothesis is consistent for every  $\mathbf{d}$ . By Theorem 11.3.2 its  $F$  statistic is

$$F = \frac{(\hat{\boldsymbol{\theta}} - \mathbf{d})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\theta}} - \mathbf{d})}{q s^2}.$$

The hypothesis implies a single-degree-of-freedom statement for every  $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$  in the family, namely  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$ . Call such a statement **Scheffé-rejected** if its hypothesized value lies

outside the Scheffé interval [Eq. 13.2.2](#).

### Corollary 13.2.3. The $F$ test and Scheffé's intervals

In the setting of [Theorem 13.2.2\(b\)](#), for every  $\mathbf{d} \in \mathbb{R}^q$ :

- (a)  $\max_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}^\top \hat{\boldsymbol{\theta}} - \mathbf{a}^\top \mathbf{d})^2}{s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}} = qF$ , attained at  $\mathbf{a} = \mathbf{W}^{-1}(\hat{\boldsymbol{\theta}} - \mathbf{d})$ ;
- (b) the level- $\alpha$   $F$  test rejects  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  iff at least one statement  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$ ,  $\boldsymbol{\lambda} = \boldsymbol{\Lambda} \mathbf{a}$ , is Scheffé-rejected;
- (c) if  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  is true, the probability that any of these statements is Scheffé-rejected is exactly  $\alpha$ . For any  $\boldsymbol{\beta}$ , the probability of Scheffé-rejecting at least one *true* statement  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$ ,  $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$ , is at most  $\alpha$ .

*Proof.* (a) For  $\boldsymbol{\lambda} = \boldsymbol{\Lambda} \mathbf{a}$ , the ratio is  $(\mathbf{a}^\top \mathbf{w})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$  with  $\mathbf{w} = \hat{\boldsymbol{\theta}} - \mathbf{d}$ , and its maximum is  $\mathbf{w}^\top \mathbf{W}^{-1} \mathbf{w} / s^2 = qF$ , attained at  $\mathbf{a} = \mathbf{W}^{-1} \mathbf{w}$ , by [Corollary 1.7.8\(c\)](#) with  $\mathbf{W}$  positive definite ([Theorem 9.2.4\(a\)](#)); this is the maximum identity of [Section 11.6](#). (b) The statement for  $\mathbf{a}$  is Scheffé-rejected iff its squared  $t$  statistic exceeds  $qF_\alpha(q, \nu)$ , so some statement is rejected iff the maximum in (a) exceeds  $qF_\alpha(q, \nu)$ , that is, iff  $F > F_\alpha(q, \nu)$ . (c) The first claim follows from (b) and the exact level of the  $F$  test ([Theorem 11.3.2](#)). The second is [Proposition 13.1.2\(a\)](#) applied to the simultaneous intervals of [Theorem 13.2.2](#). ■

When the  $F$  test rejects, *some* function in the family is Scheffé-significant, though perhaps not one of interest; conversely, no function is Scheffé-significant unless the  $F$  test rejects, so no separate protecting  $F$  test is needed. For a reduced model the same holds with  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ : by [Lemma 13.2.1](#) the  $F$  statistic of [Theorem 11.1.1](#) is the maximum over  $\mathbf{v} \in \mathcal{V}$  of  $(\mathbf{v}^\top \mathbf{Y})^2 / ((r - r_0)s^2 \|\mathbf{v}\|^2)$  ([Exercise \(B1\)](#)). Taking  $\mathcal{C}(\boldsymbol{\Lambda}) = \mathcal{C}(\mathbf{X}^\top)$ , so that  $q = r$  and the family contains the mean response  $\mathbf{x}^\top \boldsymbol{\beta}$  at every  $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$ , gives the band  $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{rF_\alpha(r, \nu)} s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$ , the Working–Hotelling band of [Theorem 12.4.1](#).

## 13.2.4 Smaller families give shorter intervals

The Scheffé multiplier depends on the family only through its dimension  $q$ , and  $qF_\alpha(q, \nu)$  is strictly increasing in  $q$  for fixed  $\nu$  and  $\alpha$ , as a  $\chi^2$  coupling shows (an exercise of [Section 11.6](#)). So one should use the smallest subspace containing every function of interest, whose dimension can be smaller than the length of a list spanning it ([Exercise \(A2\)](#)). For a few planned functions, Bonferroni's method ([Section 13.4](#)) is usually shorter still, as in [Example \(Three slopes of a crime regression\)](#); Scheffé's multiplier is earned when functions are picked after the estimates are seen.

## 13.2.5 All contrasts in a one-way layout

In the one-way model  $E(y_{kj}) = \mu + \alpha_k$  with  $g$  levels,  $n_k \geq 1$  observations at level  $k$  and  $n = \sum_k n_k$ , the estimable functions of the effects alone are the contrasts  $\psi = \sum_k c_k \mu_k$  with  $\sum_k c_k = 0$  ([Theorem 8.7.1](#)). They form a space of dimension  $g - 1$ , the test space of the hypothesis of equal means. The estimate is  $\hat{\psi} = \sum_k c_k \bar{y}_k$  with standard error  $s(\sum_k c_k^2 / n_k)^{1/2}$ , and  $\nu = n - g$ . Scheffé's intervals are

$$\sum_k c_k \bar{y}_k \pm \sqrt{(g-1)F_\alpha(g-1, n-g)} s \left( \sum_k \frac{c_k^2}{n_k} \right)^{1/2} \quad \text{for every contrast } \mathbf{c}, \quad (13.2.4)$$

with simultaneous coverage exactly  $1 - \alpha$ , balanced or not. The most significant contrast has  $c_k = n_k(\bar{y}_k - \bar{y})$  (Exercise (B1)), and its squared  $t$  statistic is  $(g-1)F$ , where  $F$  is the one-way  $F$  statistic.

### Example. Education and the perceived position of a candidate

The 1996 American National Election Study asked respondents to place Bill Clinton on a seven-point scale from extremely liberal (1) to extremely conservative (7). Group the  $n = 944$  respondents by education, coded from 1 (least) to 7 (most). The group sizes are 13, 52, 248, 187, 90, 227, 127, and the mean placements are

4.08, 3.58, 3.21, 2.95, 2.79, 2.66, 2.65.

More educated respondents place Clinton further to the left. The pooled standard deviation is  $s = 1.354$  on  $\nu = 937$  degrees of freedom, and the one-way  $F$  statistic is 7.88, with  $p$ -value  $2.7 \times 10^{-8}$ . The between-groups sum of squares is 86.72.

Having seen the means, one might ask how the two least educated groups compare with the two most educated, the contrast  $\psi = \frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$ . Its estimate is 1.176 with standard error 0.223, so  $t = 5.27$ . Because the contrast was suggested by the data, the unadjusted  $t$  interval is not legitimate. The Scheffé interval Eq. 13.2.4, with multiplier  $\sqrt{6F_{0.05}(6, 937)} = 3.557$ , is  $[0.383, 1.969]$ , and it is legitimate however the contrast was found. The most significant contrast of all has  $t = 6.88 = \sqrt{6 \times 7.88}$ , and its sum of squares is the whole between-groups sum of squares.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data

y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
between = np.sum(n_k * (means - y.mean()) ** 2)
F = between / (g - 1) / s2
print("group sizes", n_k.astype(int))
print("means", means.round(3))
print(f"s = {s:.4f} on {nu} df; F = {F:.3f}, p = {stats.f.sf(F, g - 1, nu):.2g}")
```

[Run this code online](#)

```
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)) # Scheffe multiplier,
q = g - 1
```

```

def contrast(c):
    """Estimate, standard error and sum of squares of the contrast sum_k c_k mu_k.
    """
    est = c @ means
    v = np.sum(c ** 2 / n_k)
    return est, s * np.sqrt(v), est ** 2 / v

c_max = n_k * (means - y.mean())          # the most significant
    contrast
c_lohi = np.array([0.5, 0.5, 0, 0, 0, -0.5, -0.5]) # levels 1-2 against levels
    6-7
for name, c in (("maximal", c_max), ("1-2 vs 6-7", c_lohi)):
    est, se, ss = contrast(c)
    print(f"{name:11s} estimate {est:8.3f}  t = {est / se:6.2f}  SS = {ss:8.2f}"
          f"  Scheffe interval [{est - c_sch * se:.3f}, {est + c_sch * se:.3f}]")
print(f"between-groups SS = {between:.2f};  Scheffe multiplier = {c_sch:.3f}")

```

[Run this code online](#)

## 13.2.6 Exercises

### 13.2.6.1 A. Check your understanding

#### Exercise. A1

Show that for  $q = 1$  Scheffé's interval [Eq. 13.2.2](#) is the  $t$  interval of Theorem 12.1.1, and explain why no multiplicity adjustment appears.

#### Exercise. A2

In [Example \(Education and the perceived position of a candidate\)](#), an analyst decides before seeing the data to examine the three contrasts  $\frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$ ,  $\mu_1 - \mu_7$  and  $\mu_2 - \mu_6$ . Show that they span a space of dimension 2, and state the Scheffé multiplier for this family.

### 13.2.6.2 B. Practice

#### Exercise. B1

Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with ranks  $r_0 < r$ , and  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ . Show that a vector  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  lies in  $\mathcal{V}$  iff  $\mathbf{v}^\top \boldsymbol{\mu} = 0$  for every  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_0)$ . Show that the maximum over  $\mathbf{v} \in \mathcal{V}$  of the squared  $t$  statistic for  $\mathbf{v}^\top \boldsymbol{\mu} = 0$  equals  $(r - r_0)F$ , where  $F$  is the statistic of Theorem 11.1.1.

### 13.2.6.3 C. Going deeper

#### Exercise. C1

Show that Scheffé's multiplier cannot be reduced: if  $c < \sqrt{qF_\alpha(q, \nu)}$ , the intervals  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm c \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ ,  $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$ , have simultaneous coverage less than  $1 - \alpha$ . So among intervals of the form estimate  $\pm$  constant  $\times$  standard error, valid for the whole family, Scheffé's are the shortest.

*Solution.* By [Eq. 13.2.3](#), all intervals cover iff the maximal squared  $t$  statistic, which has  $q$  times an  $F(q, \nu)$  distribution, is at most  $c^2$ . So the coverage is  $\Pr\{F(q, \nu) \leq c^2/q\}$ , which is less than  $1 - \alpha$  when  $c^2/q < F_\alpha(q, \nu)$ , because the  $F$  distribution has a positive density.